

Scenarios:

1. With phylogenetic model importation

Hypothesis: the isolation probability is much higher after March 2023, what FOI should we use?

- $P(\text{isolation} \mid \text{symp, aware}) = 0.2 \rightarrow 0.5$
- FOI = ?

		<i>P(Isolation / diagnosed, symptomatic)</i>					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

Hypothesis: FOI back to the pre-August 2022 level, = 2.2, What should $P(\text{isolation} \mid \text{symp, aware})$ be?

- Fix FOI = 2.2
- $P(\text{isolation} \mid \text{symp, aware}) = ?$

		<i>P(Isolation / diagnosed, symptomatic)</i>					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

Hypothesis: Can behavior change in months where we see most of the importations prevent increase in cases?

- For week 4 – 16 (Apr-Jun) and week 26 – 34 (Sept-Oct), FOI = 0.7 (S11)

- For week 4 – 16 (Apr-Jun), FOI = 0.7 (S15)
- For week 26 – 34 (Sept-Oct), FOI = 0.7 (S16)
- FOI = 2.2 for the other time periods

Hypothesis: Can behavior change with the same number of months as in the previous hypothesis prevent increase in cases?

- Randomly select 22 weeks with FOI = 0.7 - full model period (S17)
- Randomly select 13 weeks with FOI = 0.7 - full model period (S18)
- Randomly select 9 weeks with FOI = 0.7 - full model period (S19)
- Randomly select 22 weeks with FOI = 0.7 – model period in 2023 (S23)
- Randomly select 13 weeks with FOI = 0.7 - model period in 2023 (S24)
- Randomly select 9 weeks with FOI = 0.7 - model period in 2023 (S25)
- FOI = 2.2 for the other time periods

Hypothesis: P(ISO) remains the same as the previous levels, what should the FOI be?

		<i>P(Isolation diagnosed, symptomatic)</i>					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

		<i>P(Isolation diagnosed, symptomatic)</i>					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.6		S20			S2	
	1.8		S21				

	2.0		S22				
	2.2	S7	S6	S5	S4	S3	S14

2. Without phylogenetic model importation

- FOI = 0.7, P(ISO) = 0.2, no importation (S0)
- FOI = 2.2, P(ISO) = 0.2, no importation (S26)
- FOI = 0.7, P(ISO) = 0.2, constant importation 5/10/15 every week (S8, S9, S10)

File names and meaning:

- **new_mpox2024_S#**: Baseline scenario S#.
- **0.5Xnew_mpox2024_S#**: Same as baseline scenarios but all specified 50% of phylogenetic importations.
- **2Xnew_mpox2024_S#**: Same as baseline scenarios but all specified doubling of phylogenetic importations.
- **Import_undiag_mpox2024_S#**: Same as baseline scenarios but phylogenetic importations are imported as undiagnosed (as the default is importations are diagnosed).
- **veX_mpox2024_S#**: Vaccine effectiveness wanes to X the baseline VE profile (e.g., ve0.25 = vaccine effectiveness wanes to 25% of baseline after a year of uptake) for scenario S#. Default setting X = 0.5.